🩺 DIABETES ANALYSIS DASHBOARD

PRESENTED BY
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OUTLINE

- **Problem Statement**
- **Proposed System/Solution**
- **System Development Approach**
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
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PROBLEM STATEMENT

Diabetes is a rapidly increasing health concern worldwide. Early detection and analysis of risk factors such as glucose level, BMI, age, blood pressure, and insulin levels are critical for prevention and treatment.

However, raw healthcare data is often complex and difficult to interpret. There is a need for an interactive system that can:

- Analyze patient data efficiently
- Identify high-risk individuals
- Provide meaningful insights for decision-making

PROPOSED SOLUTION

The proposed system is an interactive data analytics dashboard that:

- Visualizes diabetes-related health data
- Identifies patterns and trends among patients
- Segments patients based on risk levels
- Enables filtering and drill-down analysis
- The system helps healthcare professionals and analysts make **data-driven decisions** quickly.

SYSTEM APPROACH

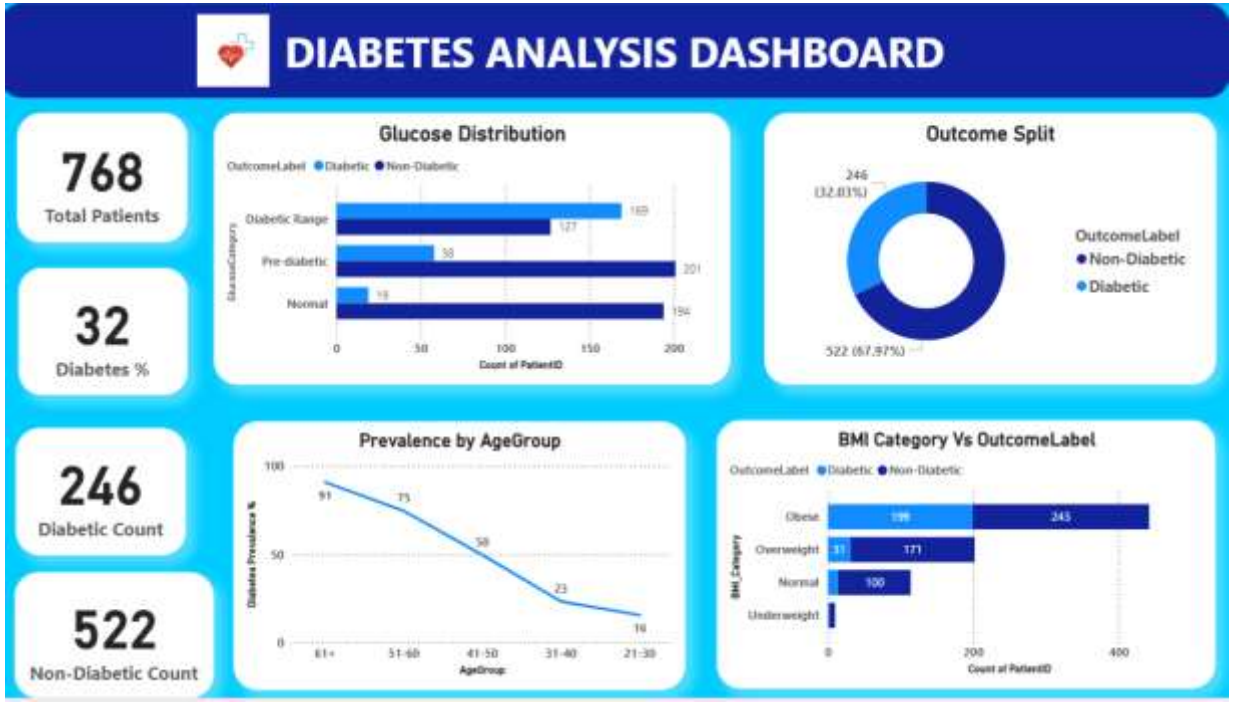
Tools & Technologies:

- Power BI → Data Visualization
- SQL → Data Extraction & Querying
- Power Query → Data Transformation

ALGORITHM & DEPLOYMENT

- **Data Collection**
  Import diabetes dataset
- **Data Preprocessing**
  Handle missing values (median) & clean data
- **Feature Engineering**
  Create Age Groups, BMI Categories, Risk Levels
- **Data Analysis**
  Identify correlations & segment patients
- **Visualization**
  Build charts & add KPI metrics
- **Deployment**
  Publish dashboard (Power BI)

RESULT


DIABETES ANALYSIS DASHBOARD
768
Total Patients
32
Diabetes %
246
Diabetic Count
522
Non-Diabetic Count
Glucose Distribution
OutcomeLabel Diabetic Non-Diabetic
Diabetic Range
Pre-diabetic
Normal
Count of PatientID
Outcome Split
246 (32.03%)
522 (67.97%)
OutcomeLabel
Non-Diabetic
Diabetic
Prevalence by AgeGroup
Diabetes Prevalence %
AgeGroup
BMI Category Vs OutcomeLabel
OutcomeLabel Diabetic Non-Diabetic
Obese
Overweight
Normal
Underweight
BMI_Category
Count of PatientID

Age Group
BMI Category
Risk Level
Outcome Label
Glucose Category
69.3
Avg Blood Pressure
31.2
Avg BMI
118.7
Avg Glucose
116
Avg Insulin
Blood Pressure Ranges
Diabetic Non-Diabetic
BloodPressure (bins)
Count of BloodPressure (bins)
AgeGroup Vs Diabetes
Diabetic Non-Diabetic
Count of PatientID
31-40
21-30
41-50
51-60
61+
AgeGroup
Pregnancies Vs Diabetes
Diabetic Non-Diabetic
Count of PatientID
Pregnancies
Diabetes Prevalence %
30.00
32.0
0.00
100.00

- High glucose levels strongly indicate diabetes risk.
- Obesity (high BMI) significantly increases chances of diabetes.
- Diabetes risk increases with age, especially in adults and seniors.
- Blood pressure has a moderate but not primary impact on diabetes.
- Glucose, BMI, and age are the most important predictors.
- Most diabetic patients fall into overweight and obese categories.
- High-risk individuals have both high glucose and high BMI.
- Data preprocessing improved the accuracy and reliability of insights.
- Dataset shows clear separation between diabetic and non-diabetic groups.
- Early detection should focus on glucose levels and body weight.

CONCLUSION

The developed dashboard successfully transforms raw healthcare data into meaningful insights. It helps:

- Identify high-risk patients
- Understand key contributing factors
- Improve decision-making using data visualization
- This project demonstrates the importance of data analytics in healthcare.

FUTURE SCOPE

- Integrate Machine Learning models for prediction
- Use real-time healthcare data
- Deploy as a web application
- Add patient-level drill-through reports
- Incorporate AI-based recommendations

REFERENCES

- Pima Indians Diabetes Dataset (Kaggle / UCI Repository)
- Microsoft Power BI Documentation
- Tableau Official Documentation
- Python (Pandas, NumPy) Documentation
- Healthcare Analytics Research Papers

THANK YOU